Piano Transcription Research: Experiment Log

Project research record

Updated 8 November 2026

1 Purpose and status

This log records the reproducibility work, code changes, fine-tuning runs, held-out evaluation, inference ablations, and timing-error diagnostics for the structured-attribute piano transcription experiments in this repository. The main question is whether adding structured velocity and timing attributes to the TransKun transcription model improves transcription quality while preserving the pretrained note-recognition baseline.

Current status. The 10,000-step fine-tuning stage is the strongest run evaluated so far. Reducing the structured-attribute contribution at inference did not improve SMD scores. Matched-note timing diagnostics show small median offset bias but substantial offset errors on some notes, particularly for longer reference durations. A paired comparison on fixed MAESTRO validation windows found no demonstrated improvement over the original checkpoint; the intervals remain wide enough that this is inconclusive rather than proof of equivalence or regression. Do not select hyperparameters on SMD going forward: this 50-piece set has already been used for iterative comparisons. Reserve the untouched MAESTRO test set for a single final evaluation after model selection.

2 Evaluation conventions

- **SMD set:** 50 recordings under `datasets/SMD-piano_v2`; all evaluations below use matching 44.1 kHz stereo WAVs and MIDI references.
- **Ground-truth pedal extension:** disabled (`--noPedalExtension`).
- **Onset tolerance:** 50 ms (`--onsetTolerance 0.05`).
- **Reported SMD scores:** macro-averaged per-recording precision, recall, F1, and overlap from `transkun.computeMetrics`; the tables show F1.
- **Metric names:**
  - `note`: onset and pitch.
  - `note+offset`: onset, pitch, and note offsets.
  - `note+velocity`: onset, pitch, and velocity.
  - `note+velocity+offset`: all three.
- **Timing error sign:** residuals are reference time minus estimated time. A negative onset or offset residual means the estimate is late; a negative duration residual means the estimated duration is longer than the reference.

Unless stated otherwise, the model uses `transkun/pretrained/2.0.conf`, batch size 1, one training process, two DataLoader workers, and the prepared 44.1 kHz MAESTRO copy. The two-worker configuration succeeded after an earlier Windows paging-file failure with the first pilot attempt.

The standard SMD score command was:

```
$py = ".\.venv\Scripts\python.exe"
$estimate = ".\checkpoint\structured-attributes-10000-smd\midi"
$truth = ".\datasets\SMD-piano_v2\midi"
$result = ".\checkpoint\structured-attributes-10000-smd\metrics.json"
& $py -m transkun.computeMetrics $estimate $truth `
  --noPedalExtension --onsetTolerance 0.05 --nProcess 1 `
  --outputJSON $result
```

3 Code and reproducibility work

Commit	Work recorded
`03b23776`	Fixed single-process MIDI evaluation argument handling, onset alignment when no notes match, consistent onset tolerance, and regression tests.
`354b4c4d`	Added structured velocity, timing, and presence factors to interval transcription, configuration, documentation, and tests.
`dd035e7a`	Added compatible pretrained-weight initialization for fine-tuning the structured-attribute model.
`bfbb2953`	Improved the MAESTRO fine-tuning workflow and terminal reporting; added UTF-8 manifest handling, a finite optimizer-step limit, zero-worker DataLoader support, documentation, and a MAESTRO manifest test.

The original packaged checkpoint contains 184 state entries: 176 compatible backbone/scorer tensors transfer to the new model, four structured-attribute head tensors start from their neutral initialization, and eight old velocity/refined-offset tensors are not reused. The new head must therefore learn its attributes during fine-tuning; the original checkpoint's velocity-inclusive SMD score is not directly comparable to an untrained new head.

The MAESTRO source manifest contained 1,276 entries. The prepared training copy resampled 233 48 kHz WAVs to 44.1 kHz and hard-linked 2,319 unchanged audio/MIDI files; the source data was preserved. No-pedal-extension metadata contains 962 training, 137 validation, and 177 test entries. A real sample loaded as a 16-second, two-channel audio slice and produced gradients in the structured-attribute head.

The repository unit suite passed: **9 tests**. Coverage includes two single-process evaluation cases, UTF-8 MAESTRO manifest parsing, two pretrained-initialization cases, and four structured-attribute cases.

4 Original-model SMD reference

The original checkpoint-era model was evaluated on all 50 SMD recordings, without ground-truth pedal extension and with 50 ms onset tolerance. Three of four reproduced F1 values closely match the repository's published SMD No Ext values; note+velocity differs by about 0.008 and needs reconciliation before a formal comparison.

Metric	Reproduced F1	README SMD No Ext F1
Note onset	0.987627	0.9876
Note onset + offset	0.785080	0.7855
Note + velocity	0.953269	0.9455
Note + velocity + offset	0.758407	0.7588

The note+velocity difference is recorded rather than attributed to a specific cause; evaluation settings, checkpoint version, or aggregation may differ and should be checked before interpreting it.

5 Fine-tuning experiments

5.1 1,000-step pilot

The pilot fine-tuned from the packaged V2 checkpoint for 1,000 optimizer steps. The first attempt with two DataLoader workers stopped before learning because Windows ran out of paging-file capacity (`WinError 1455`). It was restarted using `--dataLoaderWorkers 0`; validation then completed. The final pilot checkpoint records step 1,000.

```
$py = ".\.venv\Scripts\python.exe"
$data = ".\datasets\maestro-v3.0.0-44100"
$train = "$data\metadata-no-pedal-extension\train.pickle"
$val = "$data\metadata-no-pedal-extension\val.pickle"
& $py -u -m transkun.train .\checkpoint\structured-attributes-pilot.pt ‘
  --nProcess 1 --datasetPath $data ‘
  --datasetMetaFile_train $train --datasetMetaFile_val $val ‘
  --modelConf .\transkun\pretrained\2.0.conf --nIter 1000 ‘
  --batchSize 1 --dataLoaderWorkers 0 ‘
  --pretrainedCheckpoint .\transkun\pretrained\2.0.pt
```

MAESTRO validation	Mean NLL	Precision	Recall	F1
1k pilot	111.2404	0.7851	0.7459	0.7650

Validation F1 here is the training validation's note+offset matching score. It is not the velocity-inclusive SMD score.

5.2 SMD comparison by training stage

Each later stage was initialized from the preceding stage's checkpoint with a fresh learning-rate scheduler. Thus the stages used 1,000, then 5,000, then 10,000 additional optimizer updates (16,000 stage-wise updates in total after the packaged initialization). The checkpoint step counter is local to each stage.

```
$py = ".\.venv\Scripts\python.exe"
$data = ".\datasets\maestro-v3.0.0-44100"
$train = "$data\metadata-no-pedal-extension\train.pickle"
$val = "$data\metadata-no-pedal-extension\val.pickle"
& $py -u -m transkun.train .\checkpoint\structured-attributes-5000.pt `
  --nProcess 1 --datasetPath $data `
  --datasetMetaFile_train $train --datasetMetaFile_val $val `
  --modelConf .\transkun\pretrained\2.0.conf --nIter 5000 `
  --batchSize 1 --dataLoaderWorkers 2 `
  --pretrainedCheckpoint .\checkpoint\structured-attributes-pilot.pt

& $py -u -m transkun.train .\checkpoint\structured-attributes-10000.pt `
  --nProcess 1 --datasetPath $data `
  --datasetMetaFile_train $train --datasetMetaFile_val $val `
  --modelConf .\transkun\pretrained\2.0.conf --nIter 10000 `
  --batchSize 1 --dataLoaderWorkers 2 `
  --pretrainedCheckpoint .\checkpoint\structured-attributes-5000.pt
```

SMD F1	**1k pilot**	**5k stage**	**10k stage**	**Original model**
Note onset	0.983513	0.986316	0.986832	0.987627
Note onset + offset	0.752808	0.779505	0.779944	0.785080
Note + velocity	0.786829	0.951087	0.952227	0.953269
Note + velocity + offset	0.630164	0.751711	0.752384	0.758407

The large 1k-to-5k increase in velocity-inclusive scores is expected because the structured-attribute head was newly initialized. The 5k-to-10k gains are small. Validation F1 was 0.7825 after the 5k stage and 0.7851 after the 10k stage.

Paired 5k versus 10k comparison. On per-recording note+offset F1, the 10k stage averaged +0.000439 over 5k; 24 pieces improved and 26 worsened. A piece-resampled bootstrap 95% interval was [-0.003672, +0.004996], consistent with little or no reliable per-piece gain. The other mean paired F1 deltas were +0.000517 for note onset, +0.001141 for note+velocity, and +0.000674 for note+velocity+offset. Results use the same 50 SMD pieces and should be treated as exploratory.

6 Inference-only attribute-weight ablation

Using the 10k checkpoint, the structured-attribute contribution to semi-CRF interval scoring was varied without changing model weights. The default weight is 1.0.

Weight	**Note**	**Note+offset**	**Note+velocity**	**Note+velocity+offset**
0.0	0.926784	0.715898	0.892112	0.689540
0.5	0.984944	0.774892	0.950578	0.747845
0.75	0.986230	0.778220	0.951585	0.750646
1.0	0.986832	0.779944	0.952227	0.752384

Weight 1.0 scored higher overall than 0.5 or 0.75. For note+offset F1, the paired mean differences against weight 1.0 were -0.001723 at weight 0.75 (bootstrap 95% interval [-0.002335, -0.001077])

and -0.005052 at weight 0.5 ([-0.006206, -0.003947]). Weight 1.0 also beat weight 0.0 on all 50 pieces for each reported metric. This supports retaining the default weight; it does not establish that 1.0 is optimal on unseen data.

7 Offset timing diagnostic

The saved JSON `deviations` values are not used for this diagnostic: the repository computes those pairs with 0.8 s onset and minimum-offset tolerances, which are much looser than the reported evaluation. Instead, the diagnostic reloads predicted and reference MIDI and pairs notes using `mir_eval` rules: 50 ms onset tolerance; for onset+offset matching, offset tolerance is the greater of 20% of reference duration and 50 ms. Ground-truth pedal extension is disabled. Results below are therefore conditional on matched notes, not an error distribution over missed notes.

7.1 Matched-note residuals

Residual, reference-estimate	Subset	5k	10k
Onset median	Onset+pitch matched	0.00 ms	0.00 ms
Onset mean absolute error	Onset+pitch matched	5.45 ms	5.41 ms
Offset median	Onset+pitch matched	-3.65 ms	-2.08 ms
Offset median absolute error	Onset+pitch matched	16.15 ms	16.15 ms
Offset within 50 ms	Onset+pitch matched	76.03%	76.12%
Offset median	Strict onset+offset matched	-3.13 ms	-2.60 ms
Offset median absolute error	Strict onset+offset matched	11.46 ms	11.46 ms
Duration median	Strict onset+offset matched	-4.17 ms	-3.65 ms

For onset+pitch matched notes, the 10k checkpoint had 142,991 pairs. The offset-error mean was -1.45 ms, but mean absolute error was 70.55 ms; the 5th and 95th percentiles were -140.1 ms and +156.8 ms. The small signed median therefore hides a broad, approximately balanced error tail. Negative offset residual means the estimated note ends later than the reference.

The strict onset+offset matching procedure found 114,090 matches. Separately, applying the offset threshold to the onset-only matched pairs found 29,009 offset-threshold failures (20.29%). These counts need not be exact complements because maximum bipartite assignment can select different pairs when offset constraints are enabled.

7.2 Offset failures by reference duration

Among onset+pitch matched notes, the offset failure rates varied by reference duration:

Reference duration	Pairs	Offset fail rate	Mean absolute offset error
Under 250 ms	109,747	14.51%	32.22 ms
250–500 ms	17,906	42.09%	117.61 ms
500 ms–1 s	9,243	39.68%	204.86 ms
At least 1 s	6,095	30.83%	418.89 ms

Long notes are relatively few but have much larger absolute offset errors. This points to duration-conditioned error analysis as the next research direction rather than a global onset shift or reducing the attribute weight. The duration bins and matching are descriptive; they do not prove a specific mechanism.

7.3 MAESTRO validation timing sample

To check whether the SMD pattern generalizes beyond the repeatedly used external comparison set, a separate diagnostic was run on the 137-piece MAESTRO validation split using the 10k checkpoint. One uniformly sampled, deterministic 16-second window was selected from each piece (Python random seed 20261108). Notes were evaluated only in the center 12 seconds, leaving two seconds of context on each side. Reference notes had to be strictly contained in the window; ground-truth pedal extension was disabled. Consequently this is a sampled-window diagnostic, not full-split validation or a replacement for the training validation F1.

Among 16,500 reference notes and 15,857 estimated notes in the evaluated regions, 15,776 notes matched on pitch and onset within 50 ms. Of those, 13,020 were found by the constrained onset+offset matching. Directly applying the offset criterion to the onset-matched pairs gave 2,760 failures (17.49%). The small four-pair difference between this count and the constrained match count is due to the matching algorithm being able to choose different assignments when offset constraints are enabled.

Residual	Match subset	Signed median	Median abs.	Mean abs.
Onset	Onset+pitch	-0.18 ms	2.95 ms	3.82 ms
Offset	Onset+pitch	+1.13 ms	11.85 ms	58.49 ms
Offset	Constrained onset+offset	+0.38 ms	8.50 ms	14.75 ms
Duration	Constrained onset+offset	+0.29 ms	8.05 ms	14.53 ms

For onset+pitch matched notes, 80.13% of offsets were within an absolute 50 ms, although the mean absolute error was much larger than the median. The signed median remains close to zero; the principal issue is the error tail, not a consistent global timing shift.

Duration, pitch, and velocity strata. Offset failures below use the direct rule $|\Delta_{\text{offset}}| > \max(0.2\, d_{\text{ref}}, 50 \text{ ms})$ on the onset+pitch matched pairs. Intervals are descriptive and use a piece-cluster bootstrap (20,000 resamples of the 137 pieces), so notes from the same piece are not treated as independent bootstrap units.

Reference duration	Pairs	Failure rate	Piece-bootstrap 95% interval
Under 250 ms	12,809	11.69%	[9.94%, 13.69%]
250–500 ms	1,692	43.14%	[38.11%, 48.21%]
500 ms–1 s	825	42.18%	[36.43%, 48.07%]
At least 1 s	450	41.11%	[33.55%, 48.80%]

The offset failure rate also varied with reference velocity, whereas the three broad pitch groups were more similar:

Reference attribute bin	Pairs	Failure rate	Piece-bootstrap 95% interval
Velocity ≤ 40	1,105	37.10%	[32.92%, 40.99%]
Velocity 41–80	11,033	15.59%	[13.15%, 18.33%]
Velocity ≥ 81	3,638	17.32%	[14.06%, 20.85%]
Pitch ≤ 48	1,848	15.91%	[12.75%, 19.40%]
Pitch 49–72	9,446	18.15%	[15.50%, 21.04%]
Pitch ≥ 73	4,482	16.78%	[13.84%, 20.26%]

The low-velocity association is still present within duration strata: for notes shorter than 250 ms, the failure rate was 27.93% for velocity ≤ 40, 10.32% for velocities 41–80, and 12.52% for velocities ≥ 81. This suggests the low-velocity association is not explained only by longer durations in this sample. These univariate and stratified associations do not establish causality and may reflect annotation, performance, or model effects. The note-level records are retained locally at `checkpoint/maestro-val-offset-note-analysis.json`; the sampled window manifest and aggregate residuals are at `checkpoint/maestro-val-offset-analysis.json`.

8 Matched MAESTRO validation comparison

The original packaged checkpoint and the 10,000-step fine-tuned model were transcribed on the same deterministic sample windows from all 137 validation pieces. Each window is 16 seconds; only its saved central 12-second region is scored. The baseline uses the original checkpoint-era model implementation (the parent of commit `354b4c4d`) and all 184 packaged checkpoint tensors loaded strictly. Both models use the same stereo 44.1 kHz audio and decoding procedure. Ground truth uses the no-pedal-extension validation metadata. Metrics are macro-averaged per piece; uncertainty is a paired bootstrap over pieces (10,000 replicates, seed 20261109). Onset tolerance is 50 ms, and offset matching uses ratio 0.2 with a 50 ms minimum tolerance.

Metric	**Original F1**	**10k F1**	**Paired Δ F1 (95% interval)**
Note onset	0.964880	0.966170	+0.001290 [-0.000906, +0.003311]
Note + offset	0.753832	0.754235	+0.000403 [-0.006670, +0.007739]
Note + velocity	0.962070	0.962087	+0.000017 [-0.002429, +0.002282]
Note + velocity + offset	0.751670	0.751068	-0.000602 [-0.007605, +0.006691]

The paired note+offset effect is small, and its interval includes both improvement and degradation. The planning discussion proposed (but did not pre-register or formally agree on) a practical success threshold of at least 0.005 absolute note+offset F1 gain, a paired interval above zero, and no more than 0.002 loss in note-onset F1. This comparison does not meet that proposed threshold. The result therefore supports **no demonstrated gain**; because only one sampled window per piece was evaluated and the uncertainty is substantial, it does not establish that the models are equivalent or that structured attributes cannot help. Do not make a final success claim from these windows. The complete per-piece results are retained at `checkpoint/maestro-baseline-vs-10000-validation.json`.

9 Decision and next steps

1. Do not extend broad fine-tuning or tune on SMD: current validation evidence does not establish a practically useful gain. 2. Run one bounded, inference-only offset-score ablation on the same 137 validation windows. The hypothesis is that moderately increasing the offset-timing and offset-presence evidence during interval decoding can improve note endpoints without changing note-onset decisions substantially. Keep model weights fixed; compare the default factor scale of 1.0 against one setting, 1.5. Do not sweep additional values. 3. Before evaluating that setting, define success as at least 0.005 absolute note+offset macro-F1 gain over scale 1.0, a paired piece-bootstrap 95% interval wholly above zero, and a note-onset lower confidence bound above -0.002. If any condition fails, stop this line without another validation tweak; the test set remains reserved.

9.1 Offset-score ablation result

The offset-only factor scale was increased from 1.0 to 1.5 during interval decoding. Model weights and attribute logits were unchanged. This setting was evaluated on the same 137 windows and compared piece-by-piece with the scale-1.0 results above; uncertainty uses 10,000 paired piece-bootstrap replicates with seed 20261110.

Metric	Scale 1.0 F1	Scale 1.5 F1	Paired Δ F1 (95% interval)
Note onset	0.966170	0.966130	-0.000040 [-0.000650, +0.000624]
Note + offset	0.754235	0.754085	-0.000150 [-0.002370, +0.001996]
Note + velocity	0.962087	0.961987	-0.000100 [-0.000734, +0.000540]
Note + velocity + offset	0.751068	0.750956	-0.000112 [-0.002279, +0.002019]

The note+offset result did not meet the pre-set gain or confidence-interval criteria. The note-onset non-inferiority guard was satisfied, but that alone is insufficient for success. This targeted inference-only ablation therefore shows no practical benefit; stop further offset-score tuning and do not use the MAESTRO test set to rescue or select this setting. The per-piece output is retained at `checkpoint/maestro-offset-factor-ablation-validation.json`.

The current research result is **no demonstrated improvement** over the original model, not proof of equivalence or a universal failure of structured attributes. Further work should require a new, independently justified hypothesis and a predeclared criterion before training; otherwise, stop here.

10 Artifact locations

Generated checkpoints, MIDI predictions, and JSON metrics are local and ignored by Git under `checkpoint/`. Important result files:

- 1k SMD metrics: `checkpoint/structured-attributes-pilot-smd/metrics.json`
- 5k SMD metrics: `checkpoint/structured-attributes-5000-smd/metrics.json`
- 10k SMD metrics: `checkpoint/structured-attributes-10000-smd/metrics.json`
- Matched original-versus-10k MAESTRO validation comparison: `checkpoint/maestro-baseline-vs-10000-validation.json`
- Targeted offset-score ablation: `checkpoint/maestro-offset-factor-ablation-validation.json`
- Weight 0 SMD metrics: `checkpoint/structured-attributes-10000-no-attribute-score-smd/metrics.json`
- Weight 0.5 SMD metrics: `checkpoint/structured-attributes-10000-weight-0.5-smd/metrics.json`
- Weight 0.75 SMD metrics: `checkpoint/structured-attributes-10000-weight-0.75-smd/metrics.json`

11 References

- User-provided OpenReview paper: `https://openreview.net/pdf?id=DGA8XbJ8FVd`
- User-provided arXiv paper: `https://arxiv.org/pdf/2404.09466`
- Repository and project-specific evaluation implementation: `transkun/Evaluation.py` and `transkun/computeMetrics.py`.